Living Inside the Black Box: Behavioral Probing and Adaptation in Mandatory Wearable Sensing

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Abstract

High-stakes institutional wearables translate behavioral data into consequential judgments while obscuring how those judgments are made. We interviewed 24 people who experienced mandatory electronic monitoring in China's community corrections system. Participants built what we term *sensor literacy under constraint*, practical risk-oriented knowledge developed through uncertainty, probing, and adaptation. Mapped system behavior sometimes restored limited flexibility; costly uncertainty contracted movement and discretionary activity beyond formal rules. Some former wearers described residual habits of calculation after device removal. We propose surfacing sensor uncertainty, providing usable documentation, and evaluating behavior after device removal.

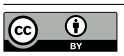
CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Empirical studies in HCI*; Empirical studies in collaborative and social computing.

Keywords

electronic monitoring, wearable sensing, sensor literacy, behavioral calibration, algorithmic opacity, design implications

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1 Introduction

Wearable sensing systems are increasingly deployed in high-stakes institutional contexts, where sensor data shape consequential decisions about people's lives [1, 25]. Electronic monitoring (EM) is an extreme instance of this condition. Individuals under community supervision wear GPS ankle devices, smart wristbands, or mobile applications continuously. These systems record location, movement, and behavioral data around the clock. The data are translated into compliance judgments with legal consequences. The interpretive logic behind those judgments remains largely inaccessible to the people wearing the devices.

In the deployment reported by participants, EM centered on location and geofence monitoring, scheduled check-ins, and device status such as battery level and signal availability. Wearers encountered the system mainly through a worn device, a paired application for some participants, and follow-up communication from supervising staff. We do not reconstruct operator-side settings or institutional decision rules. Instead, we analyze what was made legible to wearers and how they acted under those limits.

Sensor records do not carry fixed meaning. A boundary crossing, signal interruption, or timing deviation may reflect GPS drift, environmental interference, or actual non-compliance [10, 14]. Wearers rarely know how such ambiguities are resolved. They must act not only under observation, but under uncertainty about how their actions will be evaluated [13, 21].

Prior work on EM has focused on compliance outcomes, recidivism, and institutional practices [2]. It has also examined legal risks, false alerts, perceived punitiveness, and accountability around location-based supervision [11, 16, 22]. Research on wearable sensing in ubicomp and HCI has examined fitness trackers, health monitors, and other devices that users choose and can exit [5, 7]. These settings involve low stakes, user control, and voluntary participation. Much less is known about how people actively calibrate themselves to *mandatory*, continuously worn sensing systems from the wearer side. This includes how they figure out what the system records, what it tolerates, and what it does with data about

them when the system offers little explanation and little practical exit.

To address this gap, we present a qualitative study of 24 individuals who experienced EM supervision in China's community corrections system, based on semi structured interviews. We treat EM as an analytically revealing case of wearable sensing. Devices may be worn 24 hours a day. Sensed data carry immediate legal implications. System operation remains opaque to wearers. This combination makes visible a dynamic that is present but harder to observe in other wearable contexts.

Our central argument is that people experimentally learn how to live under opaque wearable sensing systems. We term this process *sensor literacy under constraint*. When wearers lack access to how their data are interpreted, they test the system in small ways. They stand near boundaries to watch the GPS dot, delay check-ins by a few minutes, and drain a device battery to measure its runtime. Over time, this learning produces orientations that vary by rule domain. In domains they had tested enough to trust, some wearers found limited working flexibility. In domains that remained uncertain, many narrowed movement and reduced discretionary activity. They stayed away from boundaries, returned home early, and avoided decisions that might brush against unknown rule edges. Some former wearers also described residual habits of caution after device removal.

This work makes the following contributions:

- We document how wearers of mandatory sensing systems actively calibrate themselves to a system they cannot inspect, through what we term *sensor literacy under constraint*. This is practical, bodily, and risk-oriented experimentation under institutional consequence. It extends prior work on folk theories by showing how probing changes when the system is continuously worn and difficult to exit.
- We characterize two orientations across rule domains. Participants reported limited flexibility where they trusted a local model of system behavior, and conservative contraction where uncertainty remained costly. Retrospective accounts from former wearers also describe residual habits of calculation after device removal.
- We derive three design implications for institutional wearable sensing systems: surfacing sensor uncertainty to wearers, providing wearer-facing documentation of system tolerances, and extending evaluation beyond the wearing period. Each implication is grounded in a specific gap between what wearers demonstrably needed and what the system provided.

EM is a high intensity case for examining wearable sensing that is difficult to refuse, consequential, remotely interpreted, and opaque to wearers. We do not claim that findings transfer directly from China's community corrections system to other settings. Instead, the case specifies a scope condition: the analysis is most relevant where wearable sensing is mandatory or hard to refuse, consequential, remotely interpreted, and opaque to wearers. The mechanisms identified here may help analyze workplace monitoring, insurance telematics, eldercare sensing, and rehabilitation tracking when those conditions are present. Understanding what these systems do to the people who wear them is a foundational question for wearable computing.

2 Related Work

Wearable sensing in everyday life. Ubiquitous computing research has long examined how sensing systems become embedded in daily routines and influence everyday behavior [1, 25]. Work on personal informatics and self-tracking has shown that wearable devices affect how people understand and regulate their own activities, from step counting to sleep monitoring [5, 7, 8]. These studies reveal that sensing systems are not merely passive recorders but active shapers of daily life. However, this body of work largely concerns voluntary, user-initiated sensing in low-stakes contexts, where individuals retain substantial control over participation, data access, and interpretation. Electronic monitoring differs in fundamental ways: wearing is compulsory, data carry immediate legal consequences, and wearers have no access to how their sensor records are interpreted or acted upon [2, 22, 24]. EM thus extends wearable sensing into a setting where sensing is institutionally enforced, consequential, and externally interpreted rather than personally managed. Our study extends this line of work by foregrounding how wearers themselves build practical knowledge when institutional sensing remains operationally opaque.

Learning to navigate sensing systems. When sensing systems operate through mechanisms that are not directly visible, users construct working understandings from indirect cues. Research on algorithmic folk theories shows that people develop informal mental models to explain system outputs in domains such as social media ranking and content recommendation [6, 9]. These models are partial and uncertain, yet they guide how people interact with and adjust their behavior in response to algorithmic systems. Related work has shown that users actively probe system behavior, experimenting with actions, observing outcomes, and revising their understanding over time [9, 12]. Surveillance and panopticism scholarship similarly shows that persistent observation can induce behavioral caution even without explicit knowledge of system rules [17].

This prior work, however, has primarily examined episodic, low-stakes interactions. A person who dislikes how their Twitter feed works can delete the app. Getting a misleading restaurant recommendation has no legal consequence. The probing documented in folk theory research happens because users are curious or frustrated, and they can walk away. In mandatory wearable sensing, the situation is different. Wearers have little practical exit. A misunderstood record can trigger a hearing. Not knowing how the system works is itself a form of vulnerability. This changes what probing means, what it costs, and what it does to people over time.

Behavioral effects after worn systems. Wearable behavior change research has documented how continuous feedback from sensing devices shapes users' activity patterns, self-perception, and decision-making over time [5, 7]. These effects are generally studied as intended outcomes of device use, such as increased physical activity or improved health awareness [8]. Less attention has been paid to behavioral residues of sensing systems that are imposed rather than chosen. When wearers lack access to how a worn system interprets their behavior, their activity can adjust toward risk reduction rather than ordinary goals. Continuous monitoring research, which shares with wearable sensing the feature of ongoing

behavioral observation, suggests that sustained observation can reorganize behavior beyond single moments of use [17]. We examine how this adaptation is reported through everyday bodily interaction with a continuously worn device [3, 27] and through former wearers' accounts after device removal.

3 Methodology

3.1 Research Design

This study uses qualitative interviews to investigate how individuals who experienced mandatory electronic monitoring calibrated themselves to a sensing system they had limited ability to inspect or exit. We focus on what people do in response to not knowing. We examine how they test the system, what they learn, how that learning shapes their behavior, and what traces they report after device removal.

3.2 Participants

Table 1: Participant demographics (U/R indicates urban/rural residence).

ID	Age	Gender	U/R	Occupation	Education
P1	28	M	R	Office Clerk	Bachelor's
P2	32	F	R	Courier	High School
P3	35	F	U	Construction Worker	Middle School
P4	55	M	U	Chef	High School
P5	29	F	R	Unemployed	High School
P6	34	M	R	Package Courier	High School
P7	26	F	U	None	High School
P8	33	M	U	Office Staff	College Degree
P9	30	M	U	Street Vendor	Middle School
P10	36	M	R	Unemployed	High School
P11	27	M	U	College Student	College Student
P12	31	M	R	Office Clerk	College Degree
P13	37	F	R	Factory Worker	Middle School
P14	33	F	R	Retail Worker	High School
P15	27	M	U	Kitchen Worker	High School
P16	45	M	R	Company Employee	Bachelor's
P17	29	M	U	Ride-hailing Driver	High School
P18	44	F	U	Unemployed	Middle School
P19	35	M	U	Construction Worker	Middle School
P20	30	M	R	Delivery Rider	High School
P21	32	M	R	Unemployed	High School
P22	31	M	R	Banquet Worker	High School
P23	34	M	R	Informal Worker	Middle School
P24	34	F	U	Office Staff	College Degree

As shown in Table 1, we recruited 24 participants. The sample included 16 men and 8 women. Ages ranged from 26 to 55, with a mean of 34. Recruitment used purposive and snowball sampling. Initial access was established through community corrections institutions, including local judicial offices, and the sample was subsequently expanded through participant referrals. All participants had experienced EM supervision for at least three months by the time of interview. At the time of interview, the sample included both current wearers and former wearers who reflected on the period after device removal. We use "wearers" to refer to people who had worn or were wearing an EM device. Claims about experiences after device removal are limited to participants who described that period. Monitoring durations ranged from three months to over two years.

Most participants wore GPS ankle devices for location and geofence monitoring. Six also used a paired mobile application, three wore smart wristbands, and two used additional physiological sensors. Because location monitoring was the common function across devices, our analysis focuses on location uncertainty and device management rather than comparing device models. Participants' access to the system was limited: they described seeing device or app feedback, receiving reminders or follow-up calls, and attending check-ins, but not seeing operator dashboards, tolerance rules, or the criteria used to classify ambiguous records. Participants represented diverse occupations, education levels, and living contexts. Thirteen participants lived in rural contexts and eleven lived in urban contexts. After the 21st interview, additional interviews produced no major new codes related to uncertainty, probing, or adaptation.

In this setting, EM records are used in routine community corrections management. Location deviations, missed check ins, low battery, or signal interruptions can trigger follow up conversations, point deductions, warnings, or further review depending on the case and local practice. Participants therefore treated ambiguous records as consequential, since technical anomalies could become part of an institutional compliance judgment.

3.3 Data Collection and Analysis

Interviews were conducted one on one, lasted 45 to 65 minutes, were audio recorded with consent, and were transcribed verbatim. Interviews were conducted in Chinese. Quotations were translated into English by bilingual researchers and checked against the original transcripts to preserve meaning, especially for terms describing uncertainty, tolerance, and violation risk. The interview guide covered participants' day to day experience under monitoring, their understanding of system operation, how they learned what the system would and would not tolerate, behavioral strategies they developed, and any changes they noticed after device removal when applicable. We employed thematic analysis. Three researchers independently coded a shared subset of transcripts, then refined a shared coding framework through discussion. The framework was applied to the full dataset, with codes grouped into higher level themes. For example, uncertainty related codes covering GPS drift, boundary ambiguity, and unpredictable enforcement were consolidated under *interpretive uncertainty*, and adjustment related codes under *coping patterns*.

The research team consisted of scholars in human-computer interaction and social computing with no institutional affiliation to the corrections system. Participants were informed that the study had no connection to their supervision. The study received IRB approval from North China University of Science and Technology. Participants provided written informed consent, were free to withdraw at any time, and received compensation of 50 RMB.

4 Results

We present findings organized around two themes: how wearers actively calibrated themselves to a system they could not inspect, and what that calibration process did to their behavior during monitoring and, for some former wearers, after device removal. The starting condition for both was the same: not knowing how the system worked was not a temporary state. It persisted, and people had to act anyway.

4.1 Calibrating to an Opaque System

4.1.1 Uncertainty as the Condition for Calibration. All 24 participants described ongoing uncertainty about how their sensor data were interpreted. This is the condition that made calibration necessary. Unlike the anxiety of being watched by a person, this uncertainty concerned the gap between their actions and how those actions became data: which GPS positions would register as violations, how timing deviations would be evaluated, and which signal interruptions would trigger consequences. Behaving more carefully did not resolve it, because the problem was not their conduct but their lack of access to how the system interpreted it.

> P16: "Having my behavior inferred from data is more distressing than being directly watched by a person. When someone watches me, I know what they see. But electronic data is different. It turns me into a set of dots and lines, and then someone judges me as a person based on those dots and lines. I have no idea what they will think."

A recurring trigger for this uncertainty was discovering that sensor records did not match wearers' understanding of their own actions. This included arriving home on time but being flagged as late, or remaining near their front door but finding a boundary crossing in the record.

> P22: "Once I was just walking around outside my front door, nowhere near leaving, but later the record showed I had gone slightly outside. I had no idea what had happened. Later I gradually understood that the GPS signal drifts. But no one told me this. I found out by bumping into it myself."

This interpretive uncertainty had consequences that extended beyond the psychological. When wearers could not predict what the system would register as a violation, they also could not predict which movements and social connections were safe to maintain. Geofence boundaries did not merely constrain movement in the abstract. They severed physical connections to people and places that fell outside the permitted zone, often without warning or recourse.

> P21: "There was a step at the hospital entrance. I sat on that step and waited. My older brother came down every so often to update me: what she ate today, what the doctor said. I do not know how long I sat there."

The physical visibility of the device added a separate dimension of difficulty. Wearing a GPS ankle monitor in public meant continuous exposure to the judgments of others, independent of what the system was recording. In a separate context, P3 reflected on the cumulative weight of this exposure:

> P3: "Sometimes I think I would rather just be locked up. Wearing this on the outside makes it genuinely hard to live well, hard to face the strange looks from other people. It is truly painful. I strongly suspect that a very close friend of mine ended our relationship because of this."

4.1.2 How Wearers Tested the System. Uncertainty was not a static condition but one that wearers actively worked to reduce. When sensor records diverged from their understanding of their own actions, participants were compelled to revise their mental models of the system. Over time, many developed operational knowledge sufficient to guide daily decisions. This included an informal understanding of GPS drift margins, timing tolerances, and which behaviors were more likely to trigger alerts. All of it was constructed from personal experience rather than any official documentation.

> P11: "After using it for a while, I developed a sense of which areas were ambiguous and which were the real red lines. That sense was something I developed from living inside it."

Rather than stopping at this uncertain grasp, almost all participants described testing the system in deliberate, small-scale ways. Participants framed these tests as attempts to avoid accidental violations rather than evade supervision. Their goal was to establish a practical safety margin for ordinary compliant activity when formal rules did not specify how sensor records would be interpreted. They chose lower-stakes situations for these tests and supplemented what they learned with observations of other supervised individuals.

Spatial probing. Participants stood near geofence boundaries to observe GPS drift behavior, testing the gap between physical position and recorded position.

> P20: "I stood at the boundary without moving, just watching whether the dot on my phone would drift outside. Sometimes it drifted out a little and then drifted back. I needed to know how large the drift range was, so I could know how large my truly safe activity space actually was."

Temporal probing. Participants used lower-stakes check-in points to test the system's tolerance window, deliberately arriving early or late by small margins to observe whether deductions followed.

> P17: "I used a less important check-in time to experiment, arriving a few minutes early and then a few minutes late. I found that a few minutes' difference did not result in a deduction. But I did not dare use the important check-in point to test this. The cost would be too high."

Hardware probing. Some participants also investigated the device itself: how long the battery lasted, under what conditions the signal dropped, and what happened when charge ran low.

> P13: "One day I did nothing else, just charged the ankle device to full and then waited for the battery to run out. I simply wanted to know how many hours it could last. That information was important. You cannot rely on guessing."

Participants were careful to calibrate the risk of each test, choosing lower-consequence scenarios for experimentation and supplementing personal tests with peer observation.

> P5: "I would not test everything myself. For some things, watching others is enough. Testing things yourself carries risk. Using other people's experience is less costly."

This peer observation was sometimes organized through informal networks among supervised individuals, who exchanged observations about system behavior at check-ins and other points of contact.

> P11: "Sometimes we run into each other during check-ins and chat for a bit. Who was recently called in for a talk, what new developments there have been somewhere else. This information is very useful, far more useful than what the official sources say. What the officials say is the rules. What we discuss is how the rules actually operate in practice."

One exception is notable: P21 described being too fearful to conduct any probing, suggesting that when monitoring instills extreme fear, wearers may forgo information gathering entirely and adopt maximum conservatism from the outset.

> P21: "I do not know. I do not dare to try. If something went wrong, I am afraid of receiving a heavier penalty."

Even for those who did probe, the knowledge gained resolved uncertainty at the level of action without eliminating it at the level of feeling.

> P8: "I already know roughly where that line is, but every time I approach it my heart still races. Knowing and feeling at ease are two different things."

What these behaviors share is that across spatial, temporal, hardware, peer-observation, and non-probing accounts (P20, P17, P13, P5, P11, P21), wearers were building knowledge the system did not provide to them. They found out about GPS drift by bumping into unexpected records. They learned timing tolerances by running small experiments at lower stakes check-ins. They measured battery life by watching it drain. Participants reported receiving no wearer oriented documentation that made these tolerances usable in daily life. The knowledge had to be earned through the body, through time, and under real legal risk. We term this process *sensor literacy under constraint*. Unlike voluntary forms of sensor literacy such as self tracking practices in personal informatics or folk theories about social media algorithms, sensor literacy under constraint is oriented toward risk management under a system that is externally interpreted and difficult to exit. That distinction changes what probing costs, what it feels like, and what it produces.

4.2 Adaptation Across Rule Domains and Reported Residues

4.2.1 Two Orientations Across Rule Domains. As participants' understanding of specific rule domains stabilized, their behavioral orientations varied by domain and perceived risk.

Strategic utilization appeared in accounts from four participants (P11, P17, P18, P23). In rule domains where participants had established reliable mental models and assessed violation consequences as manageable, they sought to maximize flexibility within those boundaries. This meant mapping precise safe zones, setting proximity alerts, and building routines that preserved normal daily activity.

> P17: "I drew out the places I could go very clearly on the map, and set up a vibration alert on my phone when I was approaching the boundary. Within the safe zone I could live normally. My mind did not have to be running all the time."

Strategic utilization depended on experiential knowledge that was specific to a rule domain. Participants described this orientation only where they trusted their local model of system behavior.

Conservative contraction appeared in accounts from 20 participants. It describes a pattern in which wearers restrict their behavior beyond what rules require. It differs from ordinary compliance in its excess. Participants did not simply follow rules. They shrank their activity space to avoid proximity to uncertainty. Where uncertainty persisted or violation consequences were perceived as severe, participants kept well away from boundaries rather than approaching them. They returned well before curfews rather than on time.

> P19: "I would rather take a detour than walk down that street near the boundary. It is not that the street is prohibited. I just do not want to be close to that line. What if the system drifts? Walking ten extra minutes is nothing, but those ten minutes give me peace of mind."

These orientations were not participant types. The same participant could act strategically in a mapped domain while remaining conservative in domains where system behavior remained uncertain. Among the 20 participants showing conservative contraction, three (P1, P4, P13) also reported behavior resembling strategic utilization in specific domains they had mapped.

4.2.2 Reported Residues After Device Removal. Participants also described cognitive and behavioral residues of monitoring. During monitoring, many described continuous calculation of location, remaining time, and device status.

Cognitively, this background calculation occupied attention regardless of activity.

> P10: "When I go out, my mind is simultaneously running several tracks: what time is it, how long before I have to be back, where am I right now, how far from the boundary, how much battery is left. When you want to do something else, there is always a feeling of something pulling at you from behind."

Over time, the burden of maintaining this calculation shaped how some participants interpreted institutional intent.

> P22: "Sometimes I wonder whether they make the rules this vague on purpose. The vaguer the rules, the less you dare to move. I do not know whether this thought is correct, but I genuinely feel this way."

Behaviorally, some former wearers reported that habits of calculating permitted locations and return times remained salient after device removal.

> P14: "I notice that now when I go out, I still instinctively think about where I can go today and when I need to be back. The device has been off for nearly three months."
>
> P4: "I notice I am now more hesitant about everything than I used to be. Before, if I wanted to go somewhere, I would just go. Now, even when there are no restrictions, I still think through many things first. That way of thinking seems to have grown into me. It is not about the device anymore, it is about everything."

These retrospective accounts identify reported residues of monitoring rather than measured psychological change. They show how habits of calculation and caution could remain salient for some participants after device removal.

5 Discussion

Participants' small tests of boundaries, check-in timing, and battery life exemplify *sensor literacy under constraint*. Wearers built risk-oriented working knowledge because the system exposed their data for institutional interpretation while disclosing little about uncertainty or tolerance. Where participants had mapped a rule domain, some recovered limited flexibility; where uncertainty persisted, many stayed well inside formal limits.

This concept distinguishes sensing uncertainty from interpretive uncertainty. Wearers questioned both whether records were accurate and how ambiguous records would become institutional judgments. Formal rules alone therefore did not support daily action. Participants had to infer practical tolerances from experience. Because our evidence comes from China's community corrections system, it does not characterize all EM deployments. The concept applies most directly where sensing is difficult to refuse, consequential, remotely interpreted, and poorly explained to wearers.

Surface sensor uncertainty to wearers. Participants could not reliably distinguish sensor error from violation, and P16 described this unpredictability as more distressing than direct observation. Systems should therefore provide practical guidance for compliant action: visible accuracy estimates, clear charging expectations, explanations for ambiguous alerts, and contestation channels. Prior work shows that partial visibility can change interaction with automated systems [13, 15]; here, operational visibility also shaped physical movement.

Design documentation for wearers, not only operators. P13 timed a full charge and discharge because no usable documentation specified battery life. Similar probing shifted the learning burden from the system to the wearer. Documentation should explain sampling, battery expectations, day-to-day treatment of uncertain records, and how ambiguity is handled. It should also identify contextual evidence that may matter, including timestamps, photos, mobility constraints [23], and signal conditions. Battery levels, GPS accuracy, and synchronization status are routine in consumer wearables; institutional systems should treat such operational legibility as support for ordinary compliant use.

Account for behavioral consequences beyond the wearing period. Some former wearers described continued calculation and caution after restrictions ended, a pattern rarely captured by current evaluations [26]. Assessments should examine movement, confidence, and decision-making after device removal, not only compliance during monitoring. This question may also matter in workplaces, insurance, and healthcare [4, 18–20] when sensing is consequential, difficult to refuse, and opaque. These adjacent domains should not be assumed to reproduce community corrections.

Limitations. This cross-sectional interview study covers one institutional context in China and captures participant accounts rather than logs or clinical measures. Retrospective reports identify perceived residues, not trajectories. Without authorities, operators, or procurement records, we cannot infer institutional intent or evaluate specific interface decisions. Future work could compare wearer accounts with operator logs and institutional practices.

6 Conclusion

People under opaque, mandatory electronic monitoring learned through *sensor literacy under constraint*: testing the system and adapting behavior to inferred tolerances. Mapped rule domains enabled limited flexibility, whereas persistent uncertainty narrowed movement beyond formal rules. Some participants reported continued calculation after device removal. Consequential wearable systems should surface sensor uncertainty, provide usable wearer-facing documentation, and evaluate effects after wearing ends. These implications apply where sensing is difficult to refuse, remotely interpreted, and opaque to wearers.

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